

Nan Xiao

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Overview

Software Engineer with a background in Applied Mathematics and Statistics, experienced in building production-grade platforms, self-service products, and LLM-based AI systems. Skilled in end-to-end system design across databases, backend, distributed infrastructure, frontend web applications. Focus on turning complex business problem into scalable, reliable and auditable products.

Education

Columbia University | *Master of Arts in Statistics*

New York | Sep 2024 – Feb 2026

Core Curriculum: Generative AI using LLMs, Natural Language Processing, Advanced Machine Learning, Statistical Machine Learning, Algorithm for Data Science, Applied Data Science, Advanced Data Analysis, Linear Regression, Probability, Inference

Chinese University of Hong Kong | *Bachelor of Science in Applied Mathematics*

Shenzhen | Sep 2020 – Jul 2024

Core Curriculum: Machine Learning, Data Structures, Optimization, Stochastic Process, Mathematical Modeling, Real Analysis

Work Experience

Antares Capital LP | Software Engineer

New York | Mar 2026 – Present

- Re-architected a legacy data grid from client-side full loading to **server-side pagination**, replacing in-memory filtering with database-driven SQL queries, pagination-optimized indexing strategies, and Redis-based distributed caching. Reduced payload size from ~50MB to ~200KB and page load time from 10–30s to 1–2s, significantly improving query latency and stability under large datasets, and enabling stateless services and horizontal scalability across multiple API servers.
- Designed and built an **enterprise workflow orchestration platform end-to-end** to replace manually coordinated, multi-week operational processes with a centralized, auditable system. Owned the full stack across **MySQL database, REST APIs, micro-frontend UI**. Enable business teams to configure, launch, and track complex processes with clear ownership.
- Architected a **versioned template-instance model** that let workflow templates evolve without disrupting active process runs. Combined point-in-time schema snapshots, independent versioning, and **optimistic concurrency** controls to preserve in-flight data while safely promoting improvements back to master templates, preventing schema drift and lost updates.
- Redesigned **authorization** into a group-based, three-tier permission system across process, step, and task scope. Centralized server-side access control with explicit precedence rules. Improved least-privilege enforcement and team accountability. Generalized user groups into a reusable organization-wide capability adopted across multiple product areas.
- Built a schema-driven configuration and monitoring framework that turned workflow onboarding into a **self-service** admin task, which automatically generated grids, forms, filters from typed schemas. Added task alerts, audit history, date tracking, real-time health indicators and **email notification** backed by scalable distributed scheduling and database aggregation.
- Developed a governed **self-service Data Product** that converts cloud warehouse tables and views into business-friendly analytics platform, enabling authorized users to discover, analyze, and save reports without database knowledge. Designed the **governance and lineage model** for Data Products with validated source mappings, UUID-based contracts, role-based access, lifecycle controls, and immutable report lineage, allowing administrators to manage approved datasets without breaking existing reports or changing their underlying meaning.

Summoner | AI Engineer

New York | Apr 2025 – Dec 2025

- Owned the **research-to-product** progression by developing a focused **proof-of-concept** for stochastic multi-agent resume evaluation and generalizing those findings into a modular, production-ready **scalable machine learning evaluation and ranking system** for automated reasoning assessment with standardized logging, monitoring, and reproducibility controls.
- Architected an **LLM-driven multi-agent evaluation platform** that ingests user's context, generates domain-agnostic challenge questions with logical reasoning pathways, synthesizes AI personas, orchestrates real-time timed challenge sessions, and evaluates answers using **LLM-based logic extraction** and **embedding cosine similarity** scoring.
- Developed a **semantic evaluation engine** using GPT-4o for structured logic extraction and **OpenAI text-embedding-3-small** with cosine similarity scoring to automate partial-credit assessment. Designed a **configurable multi-factor scoring system** with dynamic weighting across reasoning complexity, difficulty tiers, and scenario variations.
- Engineered a **cost-observability AI safeguards** using tiktoken for preflight token counting; implemented real-time USD tracking and configurable budget ceilings to prevent runaway spending across concurrent LLM operations. Optimized **high throughput distributed LLM** operations using **asynchronous Python (Asyncio)** with semaphore-bounded concurrency.
- Designed an **inter-agent collaboration protocol** based on **state-machine flow engines**, enabling persona agents to autonomously request peer hints via asynchronous message queues, integrating guidance within bounded latency windows. Engineered an **extensible integration layer** using shared JSON schemas and configs for plug-and-play agent connections.
- Built a **SQL-based evaluation analytics pipeline** to collect agent telemetry and experimental data at scale, enabling **statistical diagnostics, performance benchmarking**, and iterative model optimization. Developed a Partial Least Squares regression **analytics dashboard** using cross-validated Q^2 metrics and VIP feature important scoring to identify key performance drivers across agent configurations, informing model weighting and architecture decisions.

Technical Skills

Languages: TypeScript/JavaScript, Python, SQL, Java, HTML/CSS

AI / ML: PyTorch, Transformers, RAG, LangChain, TensorFlow, Hugging Face, Vector DB, OpenAI | Pandas, NumPy, Scikit-learn

Backend & Frontend: Kafka, REST APIs, Microservices, React, Node.js, Nest.js, Zustand, Event-driven Architecture, TypeORM,

Data & Cloud: PostgreSQL, MySQL, Redis, Git, Docker, CI/CD, Azure DevOps, GCP, AWS, Environment Configuration, Datadog